GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	**Spare Parts and Interchangeability Philosophy**
Document Number	EA-BFPCL-GPMC-GGPL-FEED-OPS-PHY-002
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Spare Parts & Interchangeability Philosophy

R03	08.06.22	Issued for Review	O. Ige	M. Ekpunobi	K. Yusuff	R03
R02	22.05.22	Issued for Review	O. Ige	M. Ekpunobi	K. Yusuff	R03
R01	13.05.22	Issued for Review	O. Ige	M. Ekpunobi	K. Yusuff	R03
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	13-05-22	Issued for Review
R02	22-05-22	Re-Issued for Review
R03	08-06-22	Re-issued for Review. Response provided in Section 4 to TCE's query "EAL to clarify how the Cost Estimate for the Spares will be done during FEED Phase."

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED-OPS-PHY-002

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)
TCE.11998A-ME-6179-LS-60002-PO-1	Mandatory Spares

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Abbreviations

Abbreviation	Meaning
AG	Associated Gas
ANSI	American National Standards Institute
API	American Petroleum Institute
BBOU	Bubouwe Bou Manifold
BFPC	Brass Fertilizer & Petrochemical Company Limited
BLV	Blowdown Valve
CAO	Computer Aided Operations
ESD	Emergency Shut Down
FEED	Front End Engineering Design
FLB	Field Logistics Base
FLM	First Line Maintenance
FQI	Flow Quantity Indicator
GSPA	Gas Supply & Purchase Agreement
HP	High Pressure
HSSE	Health Safety Security and Environment
LCP	Local Control Panel
LCV	Level Control Valve
LP	Low Pressure
MSDS	Material Safety Data Sheet
MTPA	Metric Tonnes per Annum
MTPD	Metric Tonnes Per Day
NAG	Non Associated Gas
NUPRC	Nigerian Upstream Petroleum Regulatory Commission
OML	Operating Mining License
OSD	Operational Shut Down
PAS	Process Automation System

PC	Pressure Controller
PCV	Pressure Control Valve
PEFS	Process Engineering Flow Scheme
PTW	Permit to Work
SPDC	Shell Petroleum Development Company of Nigeria

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

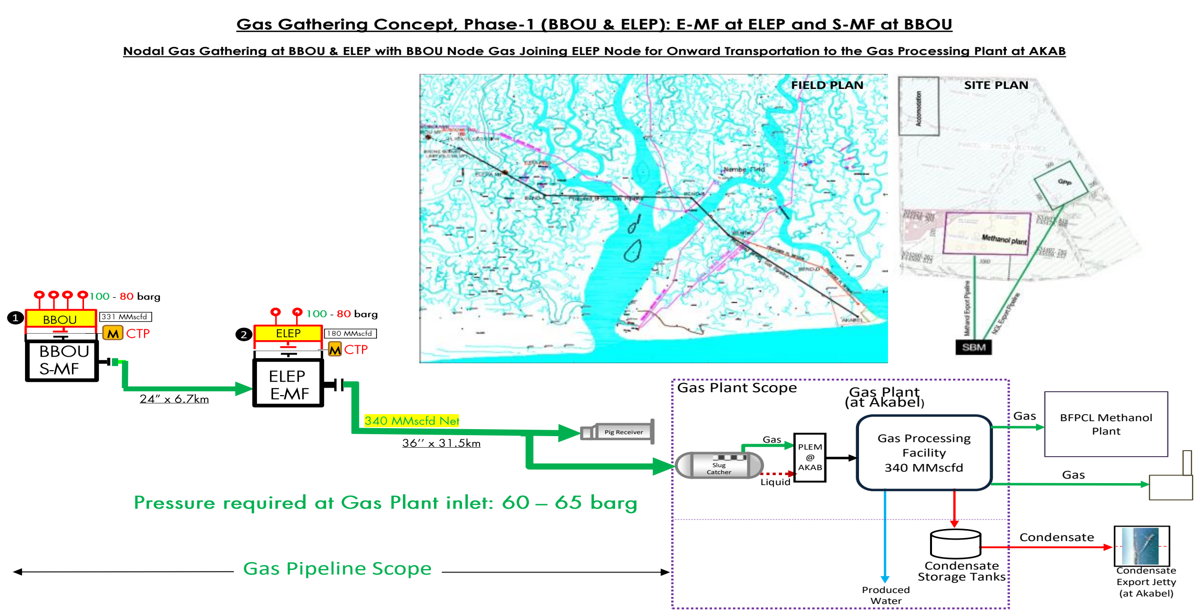

Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

To provide a framework to ensure that all spares identified and selected for this project will be properly stored and well preserved to ensure availability in optimal condition when required.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Spare Parts and Interchangeability Philosophy

This Spare Parts and Interchangeability Philosophy for Brass Fertilizer Petrochemical Company Limited (BFPCL) Gas Gathering Pipelines and Associated Infrastructure FEED defines the guidelines for selection, storage, and preservation of spare parts on this project.

The document sets out how the spare parts of this project will be managed to ensure adequacy and availability in optimum condition when required, over the full life cycle in order to enhance optimum value in accordance with the business objectives.

It will serve as a reference document to the project team and respective operations/maintenance teams in the management of the spares for this project.

This philosophy shall be updated during the detailed engineering phase of this project.

4. Inventory and Spare Parts Management

For the details of the types and quantity of spare parts required per equipment, please refer to the document:

TCE.11998A-ME-6179-LS-60002-PO-1	Mandatory Spare Parts

The list of spares will be determined during the Detailed Engineering Design when a vendor (or OEM) would have been chosen and a specific equipment selected.

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The cost estimate for the spares will be obtained as part of the bids during the EPC tendering immediately following FEED completion.

The sparing philosophy for the gas pipelines and associated infrastructure will be determined by the required availability of the gas pipelines. Due to the critical nature of the gas supply to the gas processing plant and the methanol plant, the availability will be expected to be very high (over 95%). Based on this high availability requirement, spares shall be provided for all valves, gaskets, etc. to ensure quick response and replacement of any failed material.

Spare parts holding requirements will be determined using Reliability Centred Maintenance (RCM) techniques taking into consideration replacement time and lifecycle costing for each of the components.

- All safety critical items require an all-time availability of as close to 100% as feasible. Suitable sparing and backup power systems shall ensure this level of reliability.
- Spares are categorized into three distinct categories:
  - Guarantee or Insurance spares
  - Commissioning and Start-up spares
  - Operating spares
- Critical process units and equipment shall have a sparing philosophy of at least N+1. N+2 sparing may be indicated if targets cannot be achieved with N+1. This methodology will be used for all critical equipment.
- A spare parts policy compatible with availability requirement of the facility shall be established.
- The Project team will provide Electronic Spare Parts Inter-Changeability Records (E-SPIR). Operations support and Maintenance team will review and agree to this before stocking of spares.
- Operations and Maintenance shall have the end responsibility of deciding what spare parts will be required and carried in the inventory.
- New spare parts procured must come with guaranties and full manufacturer's warranty.
- In addition to sufficient provisions for commissioning spares, routine running spares, and

insurance spares, the project team as part of the Operational readiness activities shall review and initiate if necessary, a service contract or supply chain arrangement to ensure that the required availability of any unique critical equipment will be guaranteed.

- Spare Parts and Materials will be sourced from accredited vendors and approved Original Equipment Manufacturers (OEM's) or certified dealers. Frequency of use, lead time, and cost of item will be used to determine optimal quantity of spare parts and materials to be maintained in inventory.
- The project team shall purchase all spares and ensure that the spares are fully coded prior to hand-over to Production Operations.
- Routine running spares (i.e. stock items) for all critical equipment will be stored in the warehouse and at the field base store to enhance availability.
- Spare parts for all critical equipment shall be identified and made stock items.
- Spare assemblies (insurance spares/direct charge items) will be stored in the warehouse.
- A list of INSURANCE SPARES will be agreed during detailed design prior to placing main purchase orders in line with the insurance spare policy.
- In addition to the commissioning spares, a 2-year operational spares and special maintenance tools shall be provided as part of the initial equipment purchase. These shall be available on site prior to hand-over and transferred to Production Operations.
- The project team shall replace any of the 2-year operational spares used during commissioning.
- The status of all spares shall be audited prior to hand-over from project to Operations team.

4.1. Spare Parts Procurement

The following steps shall apply;

- Equipment supplies & spares and services will be procured via the materials warehouse(s), Contracting & Procurement (C & P) Dept.
- A procurement strategy for the long-term procurement of goods and services will be determined during detailed design of facilities.
- To enhance local content, procurement of materials from local sources will be encouraged and maximised. To this end, call-off agreements, consignment stocks, and long-term contracts will be established with local suppliers wherever possible to ensure long-term support. Foreign suppliers will be considered in the absence or inadequacy of local ones.
- Material warehouses will be used for storage of materials.
- Material Supply and Inventory Management for Operations shall be managed under the C & P organizational structure.
- User departments shall submit a detailed materials forecast plan, as part of the annual Operations and Maintenance Planning process early via C & P, for the approval of BFPCL Management.

- Spare Parts and Materials will be sourced from accredited vendors and approved original equipment manufacturer (OEM) dealers.
- Warehouse services will include delivery to locations agreed with the user or customer. This will allow optimization of transport needed for delivery of goods to the fields.

4.2. Spare Parts Preservation

Spare parts shall be preserved in a conducive environment to prevent degradation and damage in order to ensure availability in optimum condition when the parts are required.

For preservation of items like valves, the valves shall be greased and inlet and outlet sealed to avoid ingress of liquid or dust into the valve bodies.

Other materials shall be kept in a dry environment shielded from weather elements to prevent rust.

Pipe openings shall be covered with a blind to prevent reptiles and pests from crawling into the pipes while in storage.